# 18.100B: Real Analysis

Your Name

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## Contents

# 1 The real numbers

## 1.1 Ordered fields and completeness

A quick reminder of notation: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are the usual number systems, and $\varepsilon > 0$ always denotes a real number.

**Definition 1.1.** Let $S \subseteq \mathbb{R}$ be nonempty. We say $b \in \mathbb{R}$ is an **upper bound** of $S$ if $x \leq b$ for every $x \in S$. The **supremum** $\sup S$ is the least upper bound of $S$, when one exists.

**Example 1.2** (Rationals are not complete)

The set $S = \{q \in \mathbb{Q} \mid q^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no least upper bound in $\mathbb{Q}$.

**Theorem 1.3** (Archimedean property)

If $x, y \in \mathbb{R}$ and $x > 0$, then there exists $n \in \mathbb{N}$ with $nx > y$.

*Proof.* Suppose not, so $A = \{nx \mid n \in \mathbb{N}\}$ is bounded above by $y$. Let $\alpha = \sup A$. Since $x > 0$, the number $\alpha - x$ is not an upper bound, so $\alpha - x < mx$ for some $m \in \mathbb{N}$. Then $\alpha < (m+1)x \in A$, contradicting $\alpha = \sup A$. □

**Corollary 1.4**

$\mathbb{Q}$ is dense in $\mathbb{R}$: if $x < y$ then there is $q \in \mathbb{Q}$ with $x < q < y$.

**Remark 1.5.** Theorem 1.3 is exactly what fails in a non-Archimedean ordered field, e.g. the field of rational functions over $\mathbb{R}$.

## 1.2 Sequences

**Definition 1.6.** A sequence $(a_n)$ **converges** to $L$ if for every $\varepsilon > 0$ there is $N$ such that $|a_n - L| < \varepsilon$ for all $n \geq N$.

**Question 1.7.** Why can a sequence not converge to two different limits?

**Exercise 1.8.** Show that every convergent sequence in $\mathbb{R}$ is bounded, and that the converse is false.

**Proposition 1.9** (Monotone convergence)

Every bounded monotone sequence of reals converges.

*Proof.* Say $(a_n)$ is increasing and bounded, and put $L = \sup_n a_n$. Given $\varepsilon > 0$, some $a_N > L - \varepsilon$; monotonicity gives

$$L - \varepsilon < a_N \leq a_n \leq L \qquad \text{for all } n \geq N,$$

hence $|a_n - L| < \varepsilon$. $\square$

## 2 Problem set

**Problem 2.1.** Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous with $f(x) = 0$ for all $x \in \mathbb{Q}$. Prove $f \equiv 0$.

*Solution.* Fix $x \in \mathbb{R}$ and take rationals $q_n \to x$. Continuity gives $f(x) = \lim_n f(q_n) = 0$. $\square$